

A Direct Factorwise Deterministic-Numerator Tail Certificate

Bin Wang

July 2026

Abstract

We derive a direct all-order target-tail bound from the parity-resolved factorization of RH-263. At the unit disk and first omitted order 29, the Fredholm trace ideal, the two even endpoint factors, and the exact odd factor give a certified logarithmic budget below 2.6624745×10^{-5} . The relative exponential error is below 2.6625100×10^{-5} . The argument uses no angular sampling and does not assert a cloud coefficient bridge or a uniform quotient theorem.

1 Tail decomposition

Let $u = z/r_H$, $r_H = 17/20$, $W = |u|^2$, and $t = |u|/\lambda$. The RH-15 factorization is

$$\log G(u) = P_1 u + \log \det(I - u^2 T) + \log A_*(u^2) + \log B(u^2) + \log C(u).$$

At $|z| = 1$, put $W = (20/17)^2$ and $t = 20/(17\lambda)$. The RH-13 certificate supplies a nuclear bound ν_1 and bounds $p_1 \geq \|T\|$, $p_2 \geq \|T^2\|$, $q \geq \|T^3\|$ as in RH-262 [1].

For $n = 3k + j$, $j \in \{1, 2, 3\}$, the trace ideal gives

$$\nu(T^n) \leq q^k \nu_j, \quad \nu_2 = p_1 \nu_1, \quad \nu_3 = p_2 \nu_1.$$

If $n \geq n_0$, then

$$\sum_{n \geq n_0} \frac{|\operatorname{tr} T^n| W^n}{n} \leq \sum_{j=1}^3 \frac{\nu_j W^j (qW^3)^{k_j}}{(3k_j + j)(1 - qW^3)}, \quad (1)$$

where k_j is the first integer with $3k_j + j \geq n_0$.

Proposition 1.1 (Endpoint tails). *For $n_0 \geq 1$ and $y = t^2 < 1$,*

$$\sum_{n \geq n_0} \frac{|[u^{2n}] \log A_*| W^n}{1} \leq \frac{y^{n_0}}{n_0(1 - y)},$$

and the corresponding B bound is this quantity divided by $2(1 - \lambda^{-2})$. For odd order $m \geq 3$ the exact coefficient is $t^m/(1 + \lambda^{-m})$ before the $1/m$ logarithmic normalization.

Proof. Use $|\lambda^{-2n}/(1 + \lambda^{-n})| \leq \lambda^{-2n}$ for A_* , $d_n \leq \lambda^{-2n}/(1 - \lambda^{-2})$ for B , and sum a geometric series after replacing each denominator by its first denominator. The odd statement is the exact RH-263 identity. $\square$

2 Certified order-29 budget

The first omitted even trace power is $n_0 = 15$ and the first omitted odd order is 29. The 100-, 150-, and 200-decimal Arb replays all satisfy the strict outward comparisons

component	certified upper bound
Fredholm even tail	0.000019045786
A_* tail	0.000003066234
B tail	0.000002376597
even total	0.000024488616
odd endpoint tail	0.000002136130
total logarithmic tail	0.000026624745
relative exponential error	0.000026625100

Theorem 2.1 (Direct factorwise target tail). *For the deterministic Hardy-scaled numerator at $|z| \leq 1$,*

$$\sum_{n \geq 29} \frac{|a_n|}{n} < 0.000026624745, \quad \exp\left(\sum_{n \geq 29} \frac{|a_n|}{n}\right) - 1 < 0.000026625100.$$

Proof. Orders $n \geq 29$ split into even $n = 2m$ with $m \geq 15$ and odd orders. Apply (1) to the Fredholm logarithm, the proposition to A_* and B , and the exact odd series to C after the $P_1 = b_1$ cancellation. The displayed outward endpoints sum to the stated bounds. $\square$

The result is deterministic and all-order in the sense of the analytic factorization; it is not an all-order theorem about noisy selected clouds. The five-component ledger remains $(0, 0, 0, 1, 1)$: no legal anchored head, cloud coefficient bridge, or uniform quotient tail has been supplied. Gates A–E remain false/open, and no Hilbert–Polya or zeta conclusion is made.

References

- [1] Bin Wang. Validated reduced-sector spectral gaps. *Preprint*, 2026. RH-13 preprint.